

Rev80 — Project Progress

Maps client meetings, stated requirements, and development phases from `VibeGui Project.md` to git commits. Useful for tracking what has been delivered and what remains.

Project Background

Originated 2025-01-23 when a Digiducer-based data collection script was demonstrated to Carmen at a client site. The goal: replace expensive commercial vibration analysis tools (e.g. VibeAnalyze, \$999) with a purpose-built Python desktop app targeting industrial predictive maintenance workflows.

Development Phases

Proto — Initial prototype (Feb – Aug 2025)

Bare-bones data capture and display using the Digiducer USB audio interface. No HDF5, no velocity spectrum, minimal GUI.

Date	Commit	Description
2025-02-15	<code>ae1736e</code>	Project created
2025-03-05	<code>5b017d1</code>	Project management notes
2025-03-17	<code>891d26f</code>	Backup snapshot
2025-05-20	<code>cca1d4e</code>	Working demo (Digiducer stream + basic plot)
2025-05-20	<code>e247dba</code>	Module init
2025-05-21	<code>856f855</code>	Digiducer abstracted into sensor and logger types
2025-08-12	<code>1257994</code>	First working dearpygui GUI with live stream
2025-08-12	<code>a3c2fd1</code>	Project management
2025-08-12	<code>be9e91c</code>	Bug fix: array dimensions

Meeting — 2025-01-23: Carmen, first demo

Requirements captured:

#	Requirement	Status
R1	Frequency range and bin size selector	✖ Done — Dec 2025
R2	Normalize FFT to IPS or g (velocity preferred over acceleration)	✖ Done — Dec 2025
R3	Display top peaks	Done — Dec 2025
R4	Smoothing	Not implemented
R5	Interactive peak inspection	Not implemented
R6	Running speed harmonics markers	Not implemented

#	Requirement	Status
R7	Bearing fault frequency markers	Not implemented
R8	Metadata: date, company, machine info, bearing types, running rate	Not implemented
R9	Trend: overall vibration over time	Done — Mar 2026
R10	Luxury: startup and coastdown trend	Future

Phase 0 — Digiducer acquisition, GUI polish (Nov – Dec 2025)

Core pipeline stabilised; units, HDF5 persistence, and modular architecture added.

Date	Commit	Description
2025-11-17	587e393	Save button added
2025-11-20	e9120d2	Logging module with YAML config; print statements removed
2025-11-25	c8bb0ae	VibeSample refactored
2025-11-26	6049f7f	Carmen meeting notes — deployment roadmap
2025-12-11	22014af	Units switching; save/load via pickle
2025-12-16	10e48a0	scipy FFT tools; GUI unit and integration controls
2025-12-17	580d254	Logger made universal
2025-12-17	383f28e	File browser fix
2025-12-17	14dd5ca	Unit conversion consistency
2025-12-17–18	772214e – 4424d3d	AcquisitionSettings upgrade (multi-step)
2025-12-29	38f22d3	Plot units upgrade — units switching throughout GUI
2025-12-29	8fb488c	refactor_util — split into sample, sensor, simulation, digiducer modules

Meeting — 2025-11-25: Carmen, deployment roadmap

Phase definitions agreed:

Phase	Scope	Budget
Phase 1	Development to date (Digiducer era)	\$2k
Phase 2	Polish and deploy to Windows	\$1k
Phase 2b	Integrate BNC DAQ for proximity probes	\$1k
Phase 3	Sales	—

Requirements from this meeting:

#	Requirement	Status
R11	Switch between velocity and acceleration in frequency domain	Done — Dec 2025 (10e48a0 , 38f22d3)

#	Requirement	Status
R12	IPS zero-to-peak trend analysis	Done — Jan 2026, extended Mar 2026
R13	RMS amplitude option	Done — Mar 2026 (5872cab , 1e040d3)
R14	Two simultaneous sensors (phasing)	Done — Mar 2026 (multi-channel pipeline)
R15	Proximity probe / DAQ support	Partially done — PicoScope replaces BNC DAQ

Meeting — 2025-12-30: Carmen

Requirements from this meeting:

#	Requirement	Status
R16	Overall vibration energy (IPS 0-P)	Done — Jan 2026 (ff4b31d)
R17	Trend plot of overall vibration over time	Done — Mar 2026 (cff702e)
R18	Fix IPS unit scaling (was ~200× wrong)	Done — Dec 2025 / confirmed Jan 2026 (e7eddde)

Phase 1b — HDF5, FFT validation, overall vibration (Jan 2026)

Date	Commit	Description
2026-01-05	dc64e37	UI_Elements moved to util for global access
2026-01-05	e3a988d	AcquisitionSettings hot-swap while streaming
2026-01-05	cde0834	Reorg and cleanup
2026-01-06	32ab828	HDF5 save/load (replaces pickle)
2026-01-09	e76fb4f	DAQ hardware evaluation notes committed
2026-01-10	f0c549e	ISO timestamp; monotonic rel_time per block
2026-01-11	e7eddde	FFT amplitude scaling validated with single-tone tests
2026-01-12	ff4b31d	Overall 0-P vibration display (acceleration and velocity)
2026-01-12	fcf9e7e	Oversample stream for better Welch FFT resolution
2026-01-16	70c51db	uldaq dependency added (MCC USB-1608FS-Plus evaluation)
2026-01-20	2f95891	MCC DAQ code examples
2026-03-09	2508756	MCC DAQ integration work

Meeting — 2026-01-11

Notes: - Overall 0-P vibration display added and demonstrated - Bin selector behaves poorly — flagged - Timestamp now FAT32-compatible (colons replaced by hyphens) - UI/UX language and jargon should match Alta's tooling - Demo screenshots needed for website

#	Requirement	Status
R19	Confirm overall vibration values are correct (vs. individual peaks)	Done — validated <code>e7eddde</code>
R20	Fix bin selector behaviour	Done — Mar 2026 (<code>a69f14f</code> , <code>3fde4bc</code>)
R21	UI/UX language aligned with Alta	Ongoing
R22	Demo screenshots for website	Not committed

Phase 2 — PicoScope 4000A acquisition backend (Mar 2026)

Hardware pivot: replace Digiducer/sounddevice with PicoScope 4000A. Enables any IEPE sensor (accelerometers, proximity probes, microphones) via standard BNC input — removes Digiducer hardware dependency.

Date	Commit	Description
2026-03-13	<code>cf89848</code>	PicoScope 4000A acquisition backend — <code>FindPicoScope()</code> , <code>PicoScopeStream</code> , ADC → mV, accumulator, USB power fallback
2026-03-13	<code>1b711c9</code>	Merge feature/picoscope into main
2026-03-13	<code>2b9aa38</code> <code>d9add87</code>	Comprehensive README added
2026-03-13	<code>91b06ce</code>	CLAUDE.md architecture guide added
2026-03-16	<code>8abb72d</code>	IEPE sensor registry, triax support, PicoScope GUI integration — <code>ScopeSensor</code> dataclass, <code>ScopeSensorRegistry</code> YAML persistence, per-channel sensitivity (mV/EU), GUI sensor library panel
2026-03-16	<code>cd40f63</code>	Harden PicoScope detection: overvoltage recovery, watchdog (5 s timeout, 3 reconnect attempts)
2026-03-16	<code>f03255d</code>	Fix handle-close between retries; extended delay after <code>PICO_NOT_RESPONDING</code>
2026-03-16	<code>67ebfd6</code>	Test: update for new default voltage range

Phase 2a — Multi-channel EU pipeline and GUI overhaul (Mar 2026)

Full multi-channel data path from hardware to display, with per-channel sensor assignment, modality-aware integration, and three-column GUI layout.

Date	Commit	Description
2026-03-18	<code>0d8911c</code>	Three-column GUI layout with primary window resizing
2026-03-18	<code>0147231</code>	Modality-aware integration (acc↔vel↔disp via FFT); mV/ EU sensitivity pipeline; unit conversion tests

Date	Commit	Description
2026-03-19	13b1095	Add mil (1/1000 in) as displacement unit
2026-03-20	afa5ce0	Fix frequency-domain integration using signed exponent
2026-03-24	09cea74	Multi-channel pipeline, scope sensor registry, EU unit refactor — PicoScopeStream multi-channel buffers; per-channel mV → EU via ScopeSensor; DataCollector fan-out
2026-03-24	4f8897c	GUI overhaul — channel config panel; per-channel status lights; dual-axis plots; scope sensor registry UI
2026-03-24	8280bd5	Test restructure: multi-channel, hardware isolation, new API coverage
2026-03-24	5d57ab4	Fix PicoScope circular buffer wraparound in streaming callback
2026-03-24	0ca1251	Replace Acquire/Configure tabs with streamlined left-panel layout
2026-03-24	c70b8f3	Visual section containers (borders, themed backgrounds) for left panel
2026-03-25	cff702e	Multi-channel data pipeline — 32-frame ring cache; ChannelResult frozen dataclass; trend store; multi-channel HDF5 layout; browse ← → buttons
2026-03-25	142dc30	Expanded multi-channel test coverage; TestFrameCache, TestTrend, TestChannelResult
2026-03-25	32410b6	Fix: overflow bitmask; default voltage range 10 → 7 (10× finer ADC resolution)
2026-03-25	23d0902	Test: overflow key handling, per-channel helpers
2026-03-25	e365099	Per-channel AC/DC coupling; results card sizing; 3 default peaks
2026-03-25	5ec3e45	Fix: reset channel config and plot series on device switch

Phase 2b — Signal generator, spectrum controls, GUI polish (Mar 2026)

AWG output for excitation testing; FFT window selection; manual axis control; left-panel redesign; siggen config persistence.

Date	Commit	Description
2026-03-25	f74dd41	Signal generator UI — Generate config tab (enable, waveform, freq, amplitude, offset); SimulatedSensor buried (CI/test only via VibeSensor.simulated())
2026-03-25	2a05ff9	FFT window type selector (Hann/Blackman-Harris/Flattop/Hamming/Boxcar/Bartlett)
2026-03-25	7e6b89f	Manual axis control — remove auto-fit on redraw; Autoscale button; Clear Cache button; full Welch frequency range (remove maxfreq display crop)
2026-03-25	ffb4121	Left panel rework — Device card, Channels card, button visibility fix (width=-1 bug), Load button restored

Date	Commit	Description
2026-03-25	83a980d	Card heights (DPG autosize_y fix); toggle button colour theming (idle/waiting/active); autoscale on first acquisition
2026-03-25	ef0b560	Channel Setup and Signal Generator buttons in Channels card
2026-03-25	d0c1068	Device card expanded with model, S/N, ID, channel count
2026-03-25	7bc729d	CH_WARNINGS_SECTION height scales to overflow channel count
2026-03-25	eb811b6	Siggen config persisted to <code>channel_assignments.yaml</code> ; restored on device connect
2026-03-25	7aa69b3	Unified VibeSample pipeline — <code>process()</code> replaces <code>get_accel()</code> + <code>fft()</code> ; returns <code>ChannelResult</code> ; cross-modality FFT integration for physically correct time-domain signal
2026-03-25	5872cab	Amplitude mode selector (RMS / 0-P / P-P) in GUI; Y-axis labels per mode
2026-03-25	1e040d3	Amplitude mode moved to <code>ScopeSensor</code> (per-sensor, not global spectrum setting)
2026-03-26	a69f14f	Modernised AcquisitionSettings — derived properties (samplerate, blocksize from maxfreq/binsize); explicit highpass/lowpass filter fields; Welch overlap; <code>n_fft_bins</code> ; <code>memory_bytes</code>
2026-03-26	ad98816	Lowpass filter added to collector pipeline
2026-03-26	3fde4bc	Spectrum tab redesign — combo presets; live-updating derived fields; filter controls
2026-03-26	36db391	Tests updated for new AcquisitionSettings API

Meeting — 2026-03-18

#	Requirement	Status
R23	Multi-channel: toggle channels on/off	Done (09cea74 , 4f8897c)
R24	Two-channel phase relationship display	Not implemented
R25	Time-domain zoom window (e.g. 5 ms)	Not implemented
R26	Track down velocity integration low-frequency noise	Partially addressed (afa5ce0); warrants further investigation
R27	Windows packaging (exe)	Not implemented

Meeting — 2026-03-25

#	Requirement	Status
R28	0-P, P-P, RMS selectable per sensor	Done (5872cab , 1e040d3)

#	Requirement	Status
R29	Acquisition settings causing long capture times	Done — AcquisitionSettings now derives samplerate / blocksize from maxfreq / binsize with correct power-of-2 arithmetic (a69f14f)
R30	Displaying wrong bins in spectrum setup	Done — Spectrum tab redesigned with live derived fields (3fde4bc)

Phase 4 — mV-domain pipeline + trend unit reflow (May 2026)

VibeSample refactored to store raw mV throughout; sensitivity conversion and Butterworth filtering moved to DataCollector.process_sample. Welch PSD cached per sample (keyed to filter/welch config); five broadband RMS overalls (integration orders -2...+2, in mV RMS) cached per sample and stored per trend timestep as a (M,5) ndarray. Unit/sensitivity/amplitude-mode changes reflow the trend at display time via DataCollector.get_trend_for_display() — no clearing required. DataCollector.process_samples() introduced as the single entry point for frame processing. GUI display loop simplified. HDF5 bumped to v4 with per-channel orders matrix; v3 files loaded in best-effort degraded mode.

Date	Commit	Description
2026-05-05	0c76c9a	refactor: mV-domain VibeSample + 5-order trend reflow
2026-05-05	5fcf09a	refactor(gui): simplify display loop via process_samples()
2026-05-05	3717286	test: update for mV-domain pipeline
2026-05-05	0a1b38c	feat(gui): configurable frame cache depth with recording-window display
2026-05-27	90a97c6	feat(picoscope): enumerate all connected scopes; hardware channel count probe

Requirements added in Phase 4

#	Source	Requirement
R28	May 2026	Configurable frame cache depth (integer selector in Acquisition config tab) with live recording-window and total-memory derived displays

Requirements Tracker

#	Source	Requirement	Status
R4	Jan 2025	Spectral smoothing	Abandoned
R5	Jan 2025	Interactive peak inspection (click peak → identify frequency)	Done — crosshairs on spectrum, time-series, and trend plots (7f9b6a8)

#	Source	Requirement	Status
R32	Aug 2026	Anti-aliasing protection — field incident: high-frequency bearing fault aliased at low power, injecting spurious spectral energy. Fixed via mandatory oversample → zero-phase digital anti-alias filter → decimate in <code>PicoScopeStream</code>, plus a widened Nyquist margin (<code>samplerate = nextpow2(2.56*maxfreq)</code>, the 2.56x convention real FFT vibration analyzers use) so the filter has real transition-band room; retired the old user-facing (off-by-default, ineffective) lowpass control. Related fixes from the same hardware investigation: a new streaming-rate watchdog flags sustained USB throughput degradation that the existing silence watchdog couldn't see, and <code>MAXFREQ_PRESETS</code> ' top three entries (100k/250k/500k Hz) were dropped — they requested sample rates far beyond this hardware's measured continuous-streaming ceiling and were already silently corrupting most captured data.	Done — see working tree (not yet committed)
R33	Aug 2026	Flexible plot layout / dockable panels — users need context-dependent layouts (e.g. spectrum + waveform vs. trend + spectrum). Investigate DPG drag-drop window docking; if not natively supported, evaluate panel-switching or split-view alternatives.	TODO
R34	Aug 2026	ISO 20816-3 harmonization — evaluation of machine vibration measured on non-rotating parts, for industrial machines >15 kW at 120–15000 rpm: the class covering the blowers and motors this tool targets. Needs velocity RMS over a <i>declared</i> 10–1000 Hz band (2–1000 Hz below 600 rpm), machine-class selection, and Zone A/B/C/D boundaries with zone-based alarming. Current gaps: the overall is not band-limited — it spans <code>highpass_fc ... fs/2</code>, up to $2.05 \times F_{\text{max}}$, measured at +25% error from content the user explicitly excluded via <code>F_max</code> — and there is no machine classification or zone display, so the app reports a velocity number the user cannot interpret without an external table.	TODO

#	Source	Requirement	Status
R35	Aug 2026	ISO 2954 instrument conformance — requirements for instruments measuring vibration severity. Needs $\pm 10\%$ amplitude accuracy over 10–1000 Hz, a measurement band that is declared and stored with the data, frequency response within tolerance at the band edges, and overload indication. Current gaps: the band is undeclared and not persisted; the 4th-order highpass at 10 Hz puts -3 dB exactly on the lower band edge rather than in the passband (corner should sit at ~ 2 –5 Hz for a 10 Hz edge, or the response be compensated). Overload detection/exclusion is already done.	TODO
R36	Aug 2026	ISO 13373-1/-2 conformance — condition monitoring: measurement procedures (-1) and processing/presentation of vibration data (-2). Substantially met already: the v4/v5 HDF5 stores the full acquisition, channel and sensor snapshot alongside the data. Remaining: a machine / measurement-point hierarchy and route concept (currently only free-text channel names and notes), and preventing or flagging a trend assembled from frames acquired under different $F_{\max}$ or band settings, which is not comparable.	TODO
R37	Aug 2026	ISO 5348 mounting guidance — mechanical mounting of accelerometers. Documentation rather than code: usable frequency range is dominated by the mount (stud $\gg$ adhesive $\gg$ magnet $\gg$ handheld probe), and a handheld probe is unusable much above ~ 1 kHz. The app will happily display a 10 kHz spectrum the mount cannot physically support, with no indication. Add mounting guidance to the README, and ideally a per-measurement mount-type field driving a usable-bandwidth warning on the spectrum.	TODO
R38	Aug 2026	Calibration traceability (ISO 16063) — ScopeSensor carries a sensitivity typed in from the sensor datasheet, with no calibration date, method, or reference standard, so a saved measurement cannot be tied to a traceable calibration. Add calibration metadata to the sensor record and persist it with each measurement. A calibrated shaker check (0.1 in/s @ 191 Hz, 100 mV/g) put the end-to-end chain within $\sim 1.6\%$, so the accuracy is there — the provenance is not.	TODO

#	Source	Requirement	Status
R39	Aug 2026	Spectral anomaly hook — fix or remove. Unwired from the GUI panel (GUI_ANOMALY_HOOK_TYPES); the hook and the headless front end are untouched. It triggers on <code>np.any(abs(spec - baseline)/baseline > threshold)</code> across every bin, but Welch runs a single segment in every shipped preset (<code>nperseg == blocksize</code>), so each noise-floor bin is chi-squared(2) with a standard deviation equal to its own mean. P(some bin of ~2000 exceeds 1.5x) is ~1.0 on healthy data, which makes <code>consecutive_n</code> a delay rather than a defence; observed firing at the earliest frame it arithmetically can. Bins 0 and 1 are additionally hard-zeroed for integration, so on any velocity/displacement channel they deviate by ~1e12 and fire permanently. A real fix is band RMS rather than per-bin, a threshold in sigma rather than fixed %, persistence on the same band across frames, and a default band of <code>highpass_fc ... maxfreq</code> — i.e. it depends on R34's declared band. Decide then: rework on top of R34, or delete the hook and its GUI/headless/config surface outright.	TODO
R40	Aug 2026	Sensor-fault detection. Disconnect a cable mid-run today and the app trends a near-zero reading as a valid healthy measurement — the machine appears to have improved. The textbook fix, reading the IEPE bias against a nominal 8–14 V window, is not available on this hardware: our coupler has a DC blocking capacitor on its output so the bias never reaches the scope on either coupling setting, and the 4000A ranges only to ± 20 V, so a 24 V supply would over-range even with a direct pre-cap tap. Software workaround (in progress): dead-channel detection from the AC signal — flag a channel whose band RMS falls below a per- ScopeSensor minimum-plausible floor, plus the open-circuit signature of a rail-ward step decaying at the coupler's high-pass time constant. Threshold to be set from measured separation between connected/still, connected/excited, shorted and open, not invented. Hardware route, if ever wanted: a coupler tap ahead of the blocking cap with a divider bringing 24 V inside ± 20 V, or a coupler exposing a fault output.	TODO
—	Future	Proximity probe support	Done — scope sensor with EU in displacement units
—	Future	Web portal for data sharing	Abandoned

#	Source	Requirement	Status
—	Future	MCC DAQ tooling (USB-1608FS-Plus)	Abandoned — PicoScope oscilloscope replaces DAQ for all current use cases

Architecture Evolution Summary

Era	Hardware	Key modules	Format
Proto (Feb–Aug 2025)	Digiducer USB audio	monolithic	—
Phase 0 (Nov–Dec 2025)	Digiducer	sensor, sample, simulation, digiducer	pickle
Phase 1b (Jan 2026)	Digiducer	+ HDF5, refactored AcquisitionSettings	HDF5
Phase 2 (Mar 2026)	PicoScope 4000A	+ picoscope, scope_sensor, scope_sensor_registry	HDF5 multi-channel
Phase 2a (Mar 2026)	PicoScope	+ ChannelResult, multi-channel pipeline, trend	HDF5 multi-channel
Phase 2b (Mar 2026)	PicoScope + AWG	+ signal generator, Welch controls, derived AcquisitionSettings	HDF5 + YAML config
Phase 4 (May 2026)	PicoScope	mV-domain VibeSample, (M,5) trend, process_samples()	HDF5 v4